

Exercise 2

Convex Functions (1 pt)

Simple functions (0.5 pt)

Are the following functions convex? Prove your statement.

1. $f(x) = x^2, x \in \mathbb{R}$
2. $f(x) = e^{x^2}, x \in \mathbb{R}$
3. $f(x, y) = x^2 + 3xy + 2y^2, x \in \mathbb{R}, y \in \mathbb{R}$

Log-sum-exp (0.5 pt)

Show that the function

$$f(\mathbf{x}) = \log(e^{x_1} + \dots + e^{x_n})$$

is convex on $\mathbb{R}^n$.

Hint: The proof is outlined in Boyd's book p. 74. Develop it yourself and provide the intermediate steps.

Geometric Mean (Bonus (0.5 pt))

Show that the geometric mean

$$f(\mathbf{x}) = \left(\prod_{i=1}^n x_i \right)^{1/n}$$

is concave on $\mathbb{R}_{++}^n$.